

NARATHIWAT, Thailand

WMO: 485830

Lat: 6.4271N

Lon: 101.8250E

Elev: 17

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	71.7	72.9	67.7	101.9	76.9	68.9	106.1	77.8	17.8	80.8	15.3	81.4	1.0	190	0.306

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.7	92.7	79.6	91.5	79.8	90.5	79.6	82.3	89.4	81.5	88.9	80.9	88.4	7.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.4	157.7	87.7	79.2	151.7	86.8	78.8	149.4	86.5	46.2	89.5	45.2	89.3	44.6	88.7	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.7	12.0	11.0	68.9	96.2	2.1	2.3	67.4	97.9	66.2	99.2	65.1	100.6	63.6	102.3		
(5)			66.3	84.2	2.3	1.4	64.6	85.2	63.3	86.0	61.9	86.8	60.3	87.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	79.2	79.9	81.3	83.0	83.7	83.1	82.5	82.5	82.0	81.3	79.9	79.2	(6)
(7)		DBStd	2.26	1.82	1.65	1.70	1.57	1.60	1.46	1.59	1.54	1.38	1.67	2.14	1.93	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11491	905	836	971	991	1044	993	1009	1006	960	971	898	907	(10)
(11)		CDD65	6016	440	416	506	541	579	543	544	541	510	506	448	442	(11)
(12)		CDH74	61343	3846	4009	5360	6434	6702	5990	5831	5658	5093	4859	3883	3678	(12)
(13)		CDH80	21560	779	1035	1833	2690	2902	2485	2321	2250	1899	1611	1002	753	(13)
(14)	Wind	WSAvg	3.2	4.0	4.2	3.8	3.6	3.1	2.7	2.5	2.8	2.7	2.5	3.8	(14)	
(15)	Precipitation	PrecAvg	106.8	8.6	2.0	3.1	2.5	5.4	5.2	5.8	6.0	8.5	11.9	24.6	23.1	(15)
(16)		PrecMax	158.1	39.8	7.1	9.1	5.7	14.9	13.1	9.6	11.5	12.4	21.7	48.9	51.1	(16)
(17)		PrecMin	69.2	0.5	0.4	0.3	0.4	1.6	1.9	1.5	2.2	4.9	4.8	12.1	6.3	(17)
(18)		PrecStd	25.6	10.2	1.7	2.7	2.0	3.5	2.7	2.6	3.1	2.4	4.5	9.7	12.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.9	88.6	90.7	93.1	93.9	93.4	93.3	93.4	92.3	91.3	89.7	88.2	(19)
(20)			MCWB	78.6	78.0	79.1	80.4	80.3	79.8	78.8	78.1	78.4	78.3	78.6	79.2	(20)
(21)		2%	DB	86.2	87.2	89.1	91.4	92.0	91.6	91.1	91.2	90.2	89.4	87.9	86.4	(21)
(22)			MCWB	77.9	77.8	78.9	80.3	80.7	79.9	79.0	78.7	78.7	78.6	78.6	78.3	(22)
(23)		5%	DB	84.8	86.1	88.0	90.2	91.0	90.3	89.7	89.7	89.1	88.0	86.5	85.2	(23)
(24)			MCWB	77.1	77.4	78.6	79.8	80.5	79.8	79.1	78.9	78.6	78.7	78.4	77.6	(24)
(25)		10%	DB	83.8	85.0	87.2	89.3	89.9	89.3	88.5	88.4	87.9	87.1	85.4	84.0	(25)
(26)			MCWB	76.4	76.9	78.3	79.6	80.1	79.6	78.9	78.7	78.5	78.5	78.1	77.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.3	80.0	80.9	82.3	83.1	82.8	81.8	81.8	82.0	82.2	81.7	81.6	(27)
(28)			MCD B	85.8	85.8	88.3	90.5	90.8	89.8	89.3	88.6	87.9	86.9	86.1	85.7	(28)
(29)		2%	WB	78.9	79.0	80.0	81.3	82.1	81.7	80.8	80.6	80.2	80.1	79.8	79.3	(29)
(30)			MCD B	84.7	85.4	87.2	89.5	90.1	89.2	88.3	88.2	87.6	86.9	85.6	84.6	(30)
(31)		5%	WB	78.0	78.2	79.4	80.7	81.3	80.9	80.0	79.9	79.6	79.3	79.0	78.3	(31)
(32)			MCD B	83.5	84.5	86.4	88.7	89.3	88.5	87.5	87.4	87.0	86.1	84.8	83.5	(32)
(33)		10%	WB	77.2	77.5	78.8	80.0	80.6	80.1	79.5	79.2	79.0	78.8	78.4	77.6	(33)
(34)			MCD B	82.4	83.6	85.8	87.7	88.5	87.7	86.9	86.7	86.4	85.6	83.8	82.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.8	11.0	12.1	13.3	13.7	13.9	13.6	14.1	13.5	12.0	9.9	9.1	(35)
(36)			MCDBR	11.4	12.5	13.4	14.4	14.8	15.0	14.8	15.3	14.9	13.7	12.7	11.5	(36)
(37)			MCWBR	5.5	5.6	6.0	6.3	6.5	6.5	6.3	6.5	6.4	6.1	5.6	5.5	(37)
(38)		5% WB	MCD BR	10.5	11.4	12.6	13.8	14.2	14.3	14.0	14.6	14.0	12.9	11.5	10.2	(38)
(39)			MCWBR	5.6	5.6	6.0	6.4	6.8	7.0	6.7	6.8	6.7	6.3	5.8	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.462	0.459	0.465	0.450	0.453	0.466	0.470	0.468	0.461	0.460	0.451	0.468	(40)	
(41)		taud	2.298	2.289	2.290	2.336	2.349	2.334	2.315	2.305	2.319	2.328	2.359	2.295	(41)	
(42)		Ebn at Noon	272	276	274	273	267	260	260	265	271	272	273	267	(42)	
(43)		Edh at Noon	43	44	44	42	40	40	41	42	42	42	40	42	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1436	1761	1841	1870	1676	1609	1591	1662	1633	1465	1256	1201	(44)	
(45)		RadStd	110	124	156	84	89	58	88	48	63	97	99	134	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.36	N/A	N/A	+0.72	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+0.31	N/A	N/A	+0.50	+0.40	+0.34	N/A	N/A	+119

Nomenclature: See separate page